

ENGR 102 Summer Bridge

Day 10 Student Quiz

Cumulative Recap Before Repetition

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace arithmetic and Boolean precedence.

4 points

```
x = 17
y = 5
result = (x // y) ** 2 + x % y
ok = result >= 11 and not (x % 2 == 0)
print(result, ok)
```

Show quotient, remainder, exponentiation, and both Boolean components before giving exact output.

Question 2. Trace an ordered decision with a missing-work condition.

4 points

```
score = 88
missing = True

if score >= 85 and not missing:
    message = "distinction"
elif score >= 70 or not missing:
    message = "pass"
else:
    message = "incomplete"

print(message)
```

Name the first matching branch and explain why the higher label is not selected.

Question 3. Repair a type mismatch before comparison.

4 points

```
raw = "12"
limit = 10
print(raw >= limit)
```

Name the actual error, explain its cause, and write a corrected comparison that evaluates numerically.

Question 4. Program construction

8 points

Write one complete Python program that satisfies every requirement.

- Read temperature as a float and calibrated as text, where exactly yes means calibrated.
- Print invalid when temperature is below -20 or above 60.
- Otherwise print ready only when calibrated is yes and temperature is from 15 through 35 inclusive; print hold for every other valid case.
- Use one if/elif/else chain and print exactly one word.

[illegible]

Test evidence / expected result